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Master's Thesis Econometrics and Management Science - Business Analytics and Quantitative
Marketing

Median-of-Means Two-Stage Least Squares Regression

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The views stated in this thesis are those of the author and not necessarily those of the supervisor, second assessor, Erasmus School of Economics or Erasmus University Rotterdam.

Abstract

Median-of-Means methods are used for instrumental variable regression and their properties are compared to the standard 2-stage least squares method.

0.1 Covering message

This is presented as a **VERY** work in progress document. There are obvious inconsistencies in style and notation, glaring omissions, and unnecessary redundancies. The structure of section 4 closely mirrors the structure of section 2, and actual content is similar to sections 2 and 3. As such, it can be skipped without losing much

1 Preliminaries

1.1 Setup and Assumptions

Let $(Y_i, X_i, Z_i)_{i=1}^n$ be an i.i.d. random sample from a joint distribution on $\mathbb{R}^3$ satisfying the structural equation

$$Y_i = \beta X_i + \varepsilon_i, \quad (1)$$

where Y_i is the variable of interest, X_i is an endogenous regressor, and Z_i is the instrumental variable (IV). Additionally, $\beta \in \mathbb{R}$ is the structural parameter of interest and ε_i is an unobserved error term.

For any random quantity f , we write $\mu_f := \mathbb{E}[f]$ and $\sigma_f^2 := \text{Var}(f)$ for its mean and variance. All population quantities below are instances of this scheme; for example, $\mu_X = \mathbb{E}[X_i]$, $\mu_{ZX} = \mathbb{E}[Z_i X_i]$, and $\sigma_{ZX}^2 = \text{Var}(Z_i X_i)$.

We impose the following conditions.

(A1) **Exogeneity.** $\mu_{Z\varepsilon} = 0$.

(A2) **Relevance.** $\mu_{ZX} \neq 0$.

(A3) **Finite second moments.** $\sigma_{ZX}^2, \sigma_{ZY}^2, \sigma_{Z\varepsilon}^2 < \infty$.

By (A1) and (A2), the structural parameter is identified as $\beta = \mu_{ZY}/\mu_{ZX}$.

Remark 1.1 (Moment conditions). *The conditions (A1)–(A3) hold for any distribution with finite second moments of the products $Z_i Y_i$, $Z_i X_i$, and $Z_i \varepsilon_i$. Thus, no assumptions are imposed on the marginal distributions of Y_i , X_i , Z_i , or ε_i .*

Remark 1.2 (Relationship between variances). *The three variances in (A3) are related. Since $Z_i Y_i = \beta Z_i X_i + Z_i \varepsilon_i$, Minkowski's inequality gives*

$$\sigma_{Z\varepsilon} \leq \sigma_{ZY} + |\beta| \sigma_{ZX}. \quad (2)$$

To see this, set $U = Z_i Y_i$ and $V = \beta Z_i X_i$, so that $U - V = Z_i \varepsilon_i$. By Minkowski's inequality for $p = 2$,

$$\sqrt{\text{Var}(U - V)} \leq \sqrt{\text{Var}(U)} + \sqrt{\text{Var}(V)},$$

which gives $\sigma_{Z\varepsilon} \leq \sigma_{ZY} + |\beta| \sigma_{ZX}$.

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1.2 Sub-Gaussian Estimators

Definition 1.3 (Sub-Gaussian Estimators). Let $(X_i)_{i=1}^n$ be a random sample with mean μ_X and variance σ_X^2 , and let $\hat{\mu}_X$ denote an estimator for μ_X . For some constant $L > 0$, $\hat{\mu}_X$ is said to be a sub-Gaussian estimator for μ_X if

$$\mathbb{P}\left[|\hat{\mu}_X - \mu_X| > L\sigma_X\sqrt{\frac{\ln(1/\delta)}{n}}\right] \leq \delta \quad \text{for all } \delta \in (0, 1). \quad (3)$$

1.3 The Median-of-Means Estimator for a Scalar Mean

We present the scalar Median-of-Means (MoM) estimator and its concentration properties. The results in this section are adapted from Devroye, Lerasle, Lugosi and Oliveira (2016) and Lugosi and Mendelson (2019).

Algorithm 1: Scalar Median-of-Means Estimator

Input: Sample $(X_i)_{i=1}^n$, number of blocks k

Output: Estimate $\hat{\mu}_{\text{MoM}}$

Partition $(X_i)_{i=1}^n$ into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$;

for $j = 1, \dots, k$ **do**

 Compute the block mean $\hat{\mu}_X^{(j)} = \frac{1}{m} \sum_{i \in B_j} X_i$;

return $\hat{\mu}_{\text{MoM}} = \text{med}(\hat{\mu}_X^{(1)}, \dots, \hat{\mu}_X^{(k)})$;

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Theorem 1.4 (MoM concentration). Let $(X_i)_{i=1}^n$ be i.i.d. with mean μ_X and variance $\sigma_X^2 < \infty$. Let k be a positive integer, $m = \lfloor n/k \rfloor$, and $\hat{\mu}_{\text{MoM}}$ the MoM estimator of μ_X . Then

$$\mathbb{P}\left[|\hat{\mu}_{\text{MoM}} - \mu_X| > \frac{2\sigma_X}{\sqrt{m}}\right] \leq e^{-k/8}. \quad (4)$$

Setting $k = \lceil 8 \ln(1/\delta) \rceil$ for $\delta \in (0, 1)$, this becomes

$$\mathbb{P}\left[|\hat{\mu}_{\text{MoM}} - \mu_X| > \sigma_X\sqrt{\frac{32 \ln(1/\delta)}{n}}\right] \leq \delta, \quad (5)$$

which is sub-Gaussian in the sense of definition 1.3 with $L = \sqrt{32}$.

Proof. The proof proceeds in three steps.

Step 1 (Chebyshev on each block). Each block mean satisfies $\mathbb{E}[\bar{X}^{(j)}] = \mu_X$ and $\text{Var}(\bar{X}^{(j)}) = \sigma_X^2/m$. By Chebyshev's inequality with threshold $2\sigma_X/\sqrt{m}$,

$$\mathbb{P}\left[|\bar{X}^{(j)} - \mu_X| > \frac{2\sigma_X}{\sqrt{m}}\right] \leq \frac{\sigma_X^2/m}{4\sigma_X^2/m} = \frac{1}{4}.$$

Step 2 (Counting argument for the median). Define $\zeta^{(j)} = \mathbf{1}\{|\bar{X}^{(j)} - \mu_X| > 2\sigma_X/\sqrt{m}\}$. By Step 1, $\mathbb{E}[\zeta^{(j)}] \leq 1/4$. Since the blocks are disjoint subsets of an i.i.d. sample, the $(\zeta^{(j)})_{j=1}^k$ are independent.

If $|\hat{\mu}_{\text{MoM}} - \mu_X| > 2\sigma_X/\sqrt{m}$, the median exceeds $\mu_X + 2\sigma_X/\sqrt{m}$ or falls below $\mu_X - 2\sigma_X/\sqrt{m}$. In either case, at least $k/2$ of the block means satisfy $|\bar{X}^{(j)} - \mu_X| > 2\sigma_X/\sqrt{m}$, since the median of

k numbers can only lie outside an interval if fewer than half the numbers lie inside it. Therefore,

$$\mathbb{P}\left[|\hat{\mu}_{\text{MoM}} - \mu_X| > \frac{2\sigma_X}{\sqrt{m}}\right] \leq \mathbb{P}\left[\sum_{j=1}^k \zeta^{(j)} \geq \frac{k}{2}\right].$$

Step 3 (Hoeffding's inequality). The sum $\sum_{j=1}^k \zeta^{(j)}$ consists of k independent random variables with $\zeta^{(j)} \in [0, 1]$ and $\mathbb{E}[\sum_j \zeta^{(j)}] \leq k/4$. For $\sum_j \zeta^{(j)}$ to reach $k/2$, it must exceed its mean by at least $k/2 - k/4 = k/4$. By Hoeffding's inequality (Hoeffding, 1963),

$$\mathbb{P}\left[\sum_{j=1}^k \zeta^{(j)} - \mathbb{E}\left[\sum_j \zeta^{(j)}\right] \geq \frac{k}{4}\right] \leq \exp\left(\frac{-2(k/4)^2}{k}\right) = e^{-k/8}.$$

This establishes (4). The sub-Gaussian form (5) follows by setting $k = \lceil 8 \ln(1/\delta) \rceil$. $\square$

2 The Standard Instrumental Variable Estimator

Definition 2.1 (Standard IV estimator). *Consider the structural equation (1). The standard IV estimator $\hat{\beta}_{\text{IV}}$ is defined by*

$$\hat{\beta}_{\text{IV}} = \frac{\hat{\mu}_{ZY}}{\hat{\mu}_{ZX}},$$

Lemma 2.2 (Error decomposition). *Under (1) and (A1),*

$$\hat{\beta}_{\text{IV}} - \beta = \frac{\hat{\mu}_{Z\varepsilon}}{\hat{\mu}_{ZX}}, \tag{6}$$

where $\mathbb{E}[\hat{\mu}_{Z\varepsilon}] = 0$.

Proof. Multiply both sides of $Y_i = \beta X_i + \varepsilon_i$ by Z_i and average:

$$\hat{\mu}_{ZY} = \frac{1}{n} \sum_{i=1}^n Z_i(\beta X_i + \varepsilon_i) = \beta \hat{\mu}_{ZX} + \hat{\mu}_{Z\varepsilon}.$$

Rearranging and dividing by $\hat{\mu}_{ZX}$ gives $\hat{\beta}_{\text{IV}} = \beta + \hat{\mu}_{Z\varepsilon}/\hat{\mu}_{ZX}$, which is (6). By (A1), $\mathbb{E}[Z_i \varepsilon_i] = 0$, so $\mathbb{E}[\hat{\mu}_{Z\varepsilon}] = 0$ by linearity of expectation. $\square$

Theorem 2.3 (Standard IV concentration). *Assume (A1)–(A3). For any $\delta \in (0, 1)$, if*

$$n \geq \frac{8\sigma_{ZX}^2}{\delta\mu_{ZX}^2}, \tag{7}$$

then

$$\mathbb{P}\left[\left|\hat{\beta}_{\text{IV}} - \beta\right| > \frac{2\sqrt{2}\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{\delta n}}\right] \leq \delta. \tag{8}$$

Proof. By Lemma 2.2, the estimation error equals the ratio $\hat{\mu}_{Z\varepsilon}/\hat{\mu}_{ZX}$. To bound this ratio we need two things simultaneously: the denominator must be bounded away from zero, and the numerator must be small. We control each requirement with a separate probability event and then combine them via the union bound.

Step 1 (Two-event decomposition). For a denominator threshold $D \in (0, |\mu_{ZX}|)$ and an error threshold $t > 0$, both to be chosen below, define

$$A(D) = \{|\hat{\mu}_{ZX}| \geq D\}, \quad B(D, t) = \{|\hat{\mu}_{Z\varepsilon}| \leq tD\}.$$

On $A(D) \cap B(D, t)$, the denominator satisfies $|\hat{\mu}_{ZX}| \geq D$ by definition of $A(D)$, so

$$\left| \hat{\beta}_{IV} - \beta \right| = \frac{|\hat{\mu}_{Z\varepsilon}|}{|\hat{\mu}_{ZX}|} \leq \frac{tD}{D} = t.$$

Hence $\mathbb{P}\left[\left|\hat{\beta}_{IV} - \beta\right| > t\right] \leq \mathbb{P}[A(D)^c] + \mathbb{P}[B(D, t)^c]$ by the union bound. For the remainder of this proof we set $D = |\mu_{ZX}|/2$; the symmetric choice that yields the cleanest constants.

Step 2 (Chebyshev bounds). We bound each failure probability in turn.

The failure event $A^c = \{|\hat{\mu}_{ZX}| < |\mu_{ZX}|/2\}$ implies a large deviation of $\hat{\mu}_{ZX}$ from its mean. Specifically, if $|\hat{\mu}_{ZX}| < |\mu_{ZX}|/2$ then, by the reverse triangle inequality,

$$|\hat{\mu}_{ZX} - \mu_{ZX}| \geq |\mu_{ZX}| - |\hat{\mu}_{ZX}| > |\mu_{ZX}| - \frac{|\mu_{ZX}|}{2} = \frac{|\mu_{ZX}|}{2}.$$

Hence $A^c \subseteq \{|\hat{\mu}_{ZX} - \mu_{ZX}| > |\mu_{ZX}|/2\}$. Since $\text{Var}(\hat{\mu}_{ZX}) = \sigma_{ZX}^2/n$, Chebyshev's inequality applied with threshold $|\mu_{ZX}|/2$ gives

$$\mathbb{P}[A^c] \leq \frac{\sigma_{ZX}^2/n}{\mu_{ZX}^2/4} = \frac{4\sigma_{ZX}^2}{n\mu_{ZX}^2}.$$

The failure event is $B^c = \{|\hat{\mu}_{Z\varepsilon}| > t|\mu_{ZX}|/2\}$. Since $\mathbb{E}[\hat{\mu}_{Z\varepsilon}] = 0$ and $\text{Var}(\hat{\mu}_{Z\varepsilon}) = \sigma_{Z\varepsilon}^2/n$, Chebyshev's inequality gives

$$\mathbb{P}[B^c] \leq \frac{\sigma_{Z\varepsilon}^2/n}{t^2\mu_{ZX}^2/4} = \frac{4\sigma_{Z\varepsilon}^2}{nt^2\mu_{ZX}^2}.$$

Step 3 (Allocate the error budget and solve for t). We require the total failure probability $\mathbb{P}[A^c] + \mathbb{P}[B^c]$ to be at most δ , and we allocate half the budget to each event.

Budget for A^c . Setting $\mathbb{P}[A^c] \leq \delta/2$ requires

$$\frac{4\sigma_{ZX}^2}{n\mu_{ZX}^2} \leq \frac{\delta}{2} \iff n \geq \frac{8\sigma_{ZX}^2}{\delta\mu_{ZX}^2},$$

which is exactly condition (7).

Setting $\mathbb{P}[B^c] \leq \delta/2$ and solving for t :

$$\frac{4\sigma_{Z\varepsilon}^2}{nt^2\mu_{ZX}^2} = \frac{\delta}{2} \implies t = \frac{2\sqrt{2}\sigma_{Z\varepsilon}}{|\mu_{ZX}|\sqrt{\delta n}}.$$

By the union bound,

$$\mathbb{P}\left[\left|\hat{\beta}_{IV} - \beta\right| > t\right] \leq \mathbb{P}[A^c] + \mathbb{P}[B^c] \leq \frac{\delta}{2} + \frac{\delta}{2} = \delta. \quad \square$$

Remark 2.4 (Polynomial dependence on δ). *The bound (8) contains the factor $1/\sqrt{\delta}$, which is polynomial in $1/\delta$. Devroye et al. (2016) show that this is unavoidable: for any estimator formed from the empirical mean of i.i.d. observations with only finite variance, the $1/\sqrt{\delta}$ rate cannot be improved without additional distributional assumptions. The MoM-based estimators in Sections 3–?? circumvent this by operating on block means.*

Remark 2.5 (The symmetric threshold is not optimal). *The choice $D = |\mu_{ZX}|/2$ was made for simplicity. It splits the distance between 0 and $|\mu_{ZX}|$ evenly, but this is not the split that minimises the error bound for given n and δ . Section 2.1 shows that treating D as a free parameter and optimising it reduces the instrument strength constant from 8 to 1 and improves the error constant by a factor of $2\sqrt{2}$.*

2.1 Optimised Threshold

The proof of Theorem 2.3 introduced a general threshold D but then fixed it symmetrically. Here we keep D free and choose it to minimise the resulting error bound symmetrically, at the cost of a more involved Step 3.

Theorem 2.6 (Standard IV concentration, optimised threshold). *Assume (A1)–(A3). Define*

$$\rho_{\text{IV}} = \frac{\sigma_{ZX}^2}{\delta n \mu_{ZX}^2}.$$

For any $\delta \in (0, 1)$, if $\rho_{\text{IV}} < 1$ (i.e., $n > \sigma_{ZX}^2/(\delta \mu_{ZX}^2)$), then the choice $D^ = (1 - \rho_{\text{IV}}^{1/3}) |\mu_{ZX}|$ is optimal and*

$$\mathbb{P} \left[\left| \hat{\beta}_{\text{IV}} - \beta \right| > \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{\delta n} (1 - \rho_{\text{IV}}^{1/3})^{3/2}} \right] \leq \delta. \quad (9)$$

Proof. Steps 1 and 2 are identical to those of Theorem 2.3 but with D left as a free parameter; we arrive at the Chebyshev bounds

$$\mathbb{P}[A(D)^c] \leq \frac{\sigma_{ZX}^2/n}{(|\mu_{ZX}| - D)^2}, \quad \mathbb{P}[B(D, t)^c] \leq \frac{\sigma_{Z\varepsilon}^2/n}{t^2 D^2}.$$

Requiring their sum to be at most δ gives the joint constraint

$$\frac{\sigma_{ZX}^2}{n(|\mu_{ZX}| - D)^2} + \frac{\sigma_{Z\varepsilon}^2}{nt^2 D^2} \leq \delta. \quad (10)$$

Step 3 (Optimise D , then solve for t). Write $D = \eta |\mu_{ZX}|$ with $\eta \in (0, 1)$. Dividing (10) through by δ and substituting $\rho_{\text{IV}} = \sigma_{ZX}^2/(\delta n \mu_{ZX}^2)$:

$$\frac{\rho_{\text{IV}}}{(1 - \eta)^2} + \frac{\sigma_{Z\varepsilon}^2}{\delta n \mu_{ZX}^2 \eta^2 t^2} \leq 1.$$

Solving for t^2 :

$$t^2 \geq \frac{\sigma_{Z\varepsilon}^2}{\delta n \mu_{ZX}^2} \cdot \frac{1}{f(\eta)}, \quad f(\eta) = \eta^2 \left(1 - \frac{\rho_{\text{IV}}}{(1 - \eta)^2} \right).$$

Minimising t is equivalent to maximising f over $\eta \in (0, 1)$. Differentiating and setting $f'(\eta) = 0$:

$$f'(\eta) = 2\eta \left(1 - \frac{\rho_{IV}}{(1-\eta)^2} \right) - \frac{2\rho_{IV}\eta^2}{(1-\eta)^3} = 0.$$

Dividing through by $2\eta > 0$:

$$1 - \frac{\rho_{IV}}{(1-\eta)^2} - \frac{\rho_{IV}\eta}{(1-\eta)^3} = 0 \iff (1-\eta)^3 - \rho_{IV}(1-\eta) - \rho_{IV}\eta = 0 \iff (1-\eta)^3 = \rho_{IV}.$$

Hence $\eta^* = 1 - \rho_{IV}^{1/3}$, which lies in $(0, 1)$ if and only if $\rho_{IV} < 1$, the stated instrument strength condition. Substituting $(1 - \eta^*)^2 = \rho_{IV}^{2/3}$:

$$f(\eta^*) = (1 - \rho_{IV}^{1/3})^2 \left(1 - \frac{\rho_{IV}}{\rho_{IV}^{2/3}} \right) = (1 - \rho_{IV}^{1/3})^2 (1 - \rho_{IV}^{1/3}) = (1 - \rho_{IV}^{1/3})^3.$$

The minimised threshold is therefore

$$t^* = \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{\delta n} (1 - \rho_{IV}^{1/3})^{3/2}},$$

and the union bound gives $\mathbb{P} \left[\left| \hat{\beta}_{IV} - \beta \right| > t^* \right] \leq \delta$. $\square$

Remark 2.7 (Comparison with the symmetric bound). As $\rho_{IV} \rightarrow 0$ (strong instruments or large n), the correction factor $(1 - \rho_{IV}^{1/3})^{-3/2} \rightarrow 1$ and (9) approaches $\sigma_{Z\varepsilon}/(|\mu_{ZX}| \sqrt{\delta n})$, a factor of $2\sqrt{2} \approx 2.83$ tighter than (8). As $\rho_{IV} \rightarrow 1$ (approaching the boundary of instrument relevance), the factor diverges, reflecting the genuine difficulty of ratio estimation near instrument irrelevance. The instrument strength condition is eight times weaker than Theorem 2.3: $\rho_{IV} < 1$ versus $\rho_{IV} < 1/8$.

2.2 Cantelli Improvement

The Chebyshev bound applied to $A(D)^c$ in the proofs above is two-sided. However, $A(D)^c = \{|\hat{\mu}_{ZX}| < D\}$ is a one-sided failure event: only a *downward* deviation of $\hat{\mu}_{ZX}$ from μ_{ZX} causes the denominator to be too small; an upward deviation makes it larger and is harmless. Cantelli's inequality exploits this one-sidedness.

Lemma 2.8 (Cantelli's inequality). *Let W be a random variable with $\mathbb{E}[W] = \mu$ and $\text{Var}(W) = \sigma^2 < \infty$. For any $\tau > 0$,*

$$\mathbb{P}[W - \mu \leq -\tau] \leq \frac{\sigma^2}{\sigma^2 + \tau^2}.$$

This is strictly smaller than the Chebyshev bound σ^2/τ^2 for all $\tau > 0$, with the greatest relative gain when $\tau \approx \sigma$.

Theorem 2.9 (Standard IV concentration, Cantelli improvement). *Assume (A1)–(A3) and $\mu_{ZX} > 0$. Write $\gamma_{IV} = \sigma_{Z\varepsilon}^2/(n\mu_{ZX}^2)$ and define*

$$h(\eta) = \eta^2 \frac{\delta(\gamma_{IV} + (1-\eta)^2) - \gamma_{IV}}{\gamma_{IV} + (1-\eta)^2}, \quad \eta \in (0, 1). \quad (11)$$

If

$$n > \frac{(1 - \delta)\sigma_{ZX}^2}{\delta\mu_{ZX}^2}, \quad \text{equivalently,} \quad \gamma_{IV} < \frac{\delta}{1 - \delta}, \quad (12)$$

then $h(\eta) > 0$ on a non-empty interval and the estimator satisfies

$$\mathbb{P}\left[\left|\hat{\beta}_{IV} - \beta\right| > \frac{\sigma_{Z\varepsilon}}{|\mu_{ZX}| \sqrt{nh(\eta_{IV}^*)}}\right] \leq \delta, \quad (13)$$

where η_{IV}^* is the maximiser of h over its feasible domain.

Proof. We use the same two-event framework with $D = \eta\mu_{ZX}$ kept general.

Step 1 (Two-event decomposition). As in Theorem 2.3, define $A(D) = \{\hat{\mu}_{ZX} \geq D\}$ and $B(D, t) = \{|\hat{\mu}_{Z\varepsilon}| \leq tD\}$. On $A(D) \cap B(D, t)$, $|\hat{\beta}_{IV} - \beta| \leq t$.

Step 2 (Cantelli on A^c , Chebyshev on B^c).

Since $\mu_{ZX} > 0$ and $D < \mu_{ZX}$, the failure event satisfies

$$A(D)^c = \{\hat{\mu}_{ZX} < D\} \subseteq \{\hat{\mu}_{ZX} - \mu_{ZX} < -(\mu_{ZX} - D)\}.$$

This is a one-sided downward deviation, so Lemma 2.8 applies with $W = \hat{\mu}_{ZX}$, $\sigma^2 = \sigma_{ZX}^2/n$, and $\tau = \mu_{ZX} - D$:

$$\mathbb{P}[A(D)^c] \leq \frac{\sigma_{ZX}^2/n}{\sigma_{ZX}^2/n + (\mu_{ZX} - D)^2}.$$

The event $B(D, t)^c = \{|\hat{\mu}_{Z\varepsilon}| > tD\}$ is two-sided (a large residual of either sign causes failure), so Chebyshev is appropriate:

$$\mathbb{P}[B(D, t)^c] \leq \frac{\sigma_{Z\varepsilon}^2/n}{t^2 D^2}.$$

Step 3 (Constraint and optimization). Setting the total failure probability to at most δ and substituting $D = \eta\mu_{ZX}$, $\gamma_{IV} = \sigma_{ZX}^2/(n\mu_{ZX}^2)$:

$$\frac{\gamma_{IV}}{\gamma_{IV} + (1 - \eta)^2} + \frac{\sigma_{Z\varepsilon}^2}{nt^2\eta^2\mu_{ZX}^2} \leq \delta.$$

Isolating the second term and solving for t^2 :

$$t^2 \geq \frac{\sigma_{Z\varepsilon}^2}{n\mu_{ZX}^2 h(\eta)},$$

where $h(\eta)$ is defined in (11). The function $h(\eta)$ is positive only when $\delta(\gamma_{IV} + (1 - \eta)^2) > \gamma_{IV}$, i.e., $(1 - \eta)^2 > \gamma_{IV}(1 - \delta)/\delta$. The feasible region is non-empty if and only if $\gamma_{IV} < \delta/(1 - \delta)$, which is condition (12). Minimizing t over the feasible region by maximizing h and applying the union bound gives (13). $\square$

Remark 2.10 (When Cantelli helps). *Compared with the optimised Chebyshev condition ($n > \sigma_{ZX}^2/(\delta\mu_{ZX}^2)$), the Cantelli condition (12) is weaker by a factor of $1/(1 - \delta)$. At $\delta = 0.05$ this ratio is $1/0.95 \approx 1.05$, so the gain is negligible for typical significance levels. The improvement is more meaningful at moderate confidence levels: at $\delta = 0.5$ the factor reaches 2. For the standard*

IV estimator the Cantelli improvement is therefore primarily of theoretical interest. By contrast, for the MoR estimator (Section ??), Cantelli extends the valid range of the instrument strength parameter from $\gamma < 1/4$ to $\gamma < 1/3$, a qualitatively more substantial gain.

Remark 2.11 (No closed form for η_{IV}^*). The first-order condition $h'(\eta) = 0$ reduces to a quartic in η with no general closed-form solution. The optimised Chebyshev version (Theorem 2.6) yielded the clean form $\eta^* = 1 - \rho_{IV}^{1/3}$ because the Chebyshev bound σ^2/τ^2 produces a simpler algebraic structure than the Cantelli bound $\sigma^2/(\sigma^2 + \tau^2)$. The bound (13) must therefore be evaluated numerically for specific parameter values.

3 The Ratio-of-Medians MoM-IV Estimator

Algorithm 2: Ratio-of-Medians Estimator

Input: Sample $(Y_i, X_i, Z_i)_{i=1}^n$, number of blocks k

Output: Estimate $\hat{\beta}_{RM}$

Partition $(Y_i, X_i, Z_i)_{i=1}^n$ into k blocks $B_1, \dots, B_k$ of size $m = \lfloor n/k \rfloor$;

for $j = 1, \dots, k$ **do**

Compute the block means $\hat{\mu}_{ZY}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i Y_i$ and $\hat{\mu}_{ZX}^{(j)} = \frac{1}{m} \sum_{i \in B_j} Z_i X_i$;

Compute coordinate-wise medians $\tilde{\mu}_{ZY} = \text{med}(\hat{\mu}_{ZY}^{(1)}, \dots, \hat{\mu}_{ZY}^{(k)})$ and

$\tilde{\mu}_{ZX} = \text{med}(\hat{\mu}_{ZX}^{(1)}, \dots, \hat{\mu}_{ZX}^{(k)})$;

return $\hat{\beta}_{RM} = \frac{\tilde{\mu}_{ZY}}{\tilde{\mu}_{ZX}}$;

Theorem 3.1 (RoM concentration). Assume (A1)–(A3). Let $\delta \in (0, 1)$, $k = \lceil 8 \ln(2/\delta) \rceil$, and $m = \lfloor n/k \rfloor$. If the instrument strength condition

$$m > \frac{4\sigma_{ZX}^2}{\mu_{ZX}^2} \quad (14)$$

holds, then

$$\mathbb{P}\left[\left|\hat{\beta}_{RM} - \beta\right| > \frac{2(\sigma_{ZY} + |\beta|\sigma_{ZX})}{|\mu_{ZX}|\sqrt{m} - 2\sigma_{ZX}}\right] \leq \delta. \quad (15)$$

Proof. The proof applies the scalar MoM result (Theorem 1.4) to each coordinate separately and combines the results with a union bound.

Step 1 (MoM on each coordinate). By Theorem 1.4 applied to the sequences $(Z_i Y_i)_{i=1}^n$ and $(Z_i X_i)_{i=1}^n$,

$$\mathbb{P}\left[|\tilde{\mu}_{ZY} - \mu_{ZY}| > \frac{2\sigma_{ZY}}{\sqrt{m}}\right] \leq e^{-k/8}, \quad \mathbb{P}\left[|\tilde{\mu}_{ZX} - \mu_{ZX}| > \frac{2\sigma_{ZX}}{\sqrt{m}}\right] \leq e^{-k/8}.$$

Step 2 (Union bound). Define the events

$$A = \left\{|\tilde{\mu}_{ZY} - \mu_{ZY}| \leq \frac{2\sigma_{ZY}}{\sqrt{m}}\right\}, \quad B = \left\{|\tilde{\mu}_{ZX} - \mu_{ZX}| \leq \frac{2\sigma_{ZX}}{\sqrt{m}}\right\}.$$

By Step 1, $\mathbb{P}[A^c] \leq e^{-k/8}$ and $\mathbb{P}[B^c] \leq e^{-k/8}$. Note that A and B are not independent (both are

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constructed from the same data), so we use the union bound:

$$\mathbb{P}[(A \cap B)^c] = \mathbb{P}[A^c \cup B^c] \leq \mathbb{P}[A^c] + \mathbb{P}[B^c] \leq 2e^{-k/8}.$$

Setting $k = \lceil 8 \ln(2/\delta) \rceil$ ensures $2e^{-k/8} \leq \delta$.

Step 3 (Algebraic bound on the ratio). On $A \cap B$, we decompose

$$\hat{\beta}_{\text{RM}} - \beta = \frac{\tilde{\mu}_{ZY}}{\tilde{\mu}_{ZX}} - \frac{\mu_{ZY}}{\mu_{ZX}} = \frac{\tilde{\mu}_{ZY}\mu_{ZX} - \mu_{ZY}\tilde{\mu}_{ZX}}{\tilde{\mu}_{ZX} \cdot \mu_{ZX}}.$$

Adding and subtracting $\mu_{ZY}\mu_{ZX}$ in the numerator and using $\mu_{ZY} = \beta\mu_{ZX}$,

$$\tilde{\mu}_{ZY}\mu_{ZX} - \mu_{ZY}\tilde{\mu}_{ZX} = \mu_{ZX}[(\tilde{\mu}_{ZY} - \mu_{ZY}) - \beta(\tilde{\mu}_{ZX} - \mu_{ZX})].$$

The factor μ_{ZX} cancels with the μ_{ZX} in the denominator, giving

$$\hat{\beta}_{\text{RM}} - \beta = \frac{(\tilde{\mu}_{ZY} - \mu_{ZY}) - \beta(\tilde{\mu}_{ZX} - \mu_{ZX})}{\tilde{\mu}_{ZX}}.$$

Taking absolute values and applying the triangle inequality to the numerator,

$$|\hat{\beta}_{\text{RM}} - \beta| \leq \frac{|\tilde{\mu}_{ZY} - \mu_{ZY}| + |\beta| |\tilde{\mu}_{ZX} - \mu_{ZX}|}{|\tilde{\mu}_{ZX}|}.$$

Step 4 (Substitute bounds). On event A , $|\tilde{\mu}_{ZY} - \mu_{ZY}| \leq 2\sigma_{ZY}/\sqrt{m}$. On event B , $|\tilde{\mu}_{ZX} - \mu_{ZX}| \leq 2\sigma_{ZX}/\sqrt{m}$, and by the reverse triangle inequality,

$$|\tilde{\mu}_{ZX}| \geq |\mu_{ZX}| - |\tilde{\mu}_{ZX} - \mu_{ZX}| \geq |\mu_{ZX}| - \frac{2\sigma_{ZX}}{\sqrt{m}}.$$

The instrument strength condition (14) ensures $2\sigma_{ZX}/\sqrt{m} < |\mu_{ZX}|$, so the denominator is strictly positive. Substituting,

$$|\hat{\beta}_{\text{RM}} - \beta| \leq \frac{2\sigma_{ZY}/\sqrt{m} + |\beta| \cdot 2\sigma_{ZX}/\sqrt{m}}{|\mu_{ZX}| - 2\sigma_{ZX}/\sqrt{m}} = \frac{2(\sigma_{ZY} + |\beta|\sigma_{ZX})}{|\mu_{ZX}| \sqrt{m} - 2\sigma_{ZX}}.$$

Since this holds on $A \cap B$ and $\mathbb{P}[(A \cap B)^c] \leq \delta$, the result follows. $\square$

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